

3.1 Employees

employees

- id (UUID, PK)
- employee_code (UNIQUE)
- first_name
- last_name
- email (UNIQUE)
- phone
- date_of_birth
- hire_date
- employment_status (active, terminated, probation)
- department_id (FK → departments.id)
- position_id (FK → positions.id)
- manager_id (FK → employees.id, self-reference)
- created_at
- updated_at
- deleted_at

Purpose

Central entity. Self-reference enables hierarchy (manager system).

3.2 Departments

departments

- id (UUID, PK)
- name (UNIQUE)
- parent_department_id (FK → departments.id)
- created_at
- updated_at

Purpose

Supports organizational hierarchy (e.g., Engineering → Backend Team).

3.3 Positions / Roles

positions

- id (UUID, PK)
- title

- level (junior, mid, senior)
 - created_at
 - updated_at
-

3.4 Employment Details (Separation for flexibility)

employee_details

- id (UUID, PK)
 - employee_id (FK → employees.id)
 - address
 - emergency_contact
 - marital_status
 - nationality
-

4. Attendance Module

4.1 Attendance Records

attendance

- id (UUID, PK)
 - employee_id (FK)
 - date
 - check_in_time
 - check_out_time
 - status (present, absent, late, half-day)
 - created_at
-

4.2 Work Schedule

work_schedules

- id (UUID, PK)
- employee_id (FK)
- shift_start
- shift_end
- working_days (JSON / array)

5. Leave Management

5.1 Leave Types

leave_types

- id (UUID, PK)
 - name (annual, sick, unpaid)
 - max_days_per_year
-

5.2 Leave Requests

leave_requests

- id (UUID, PK)
 - employee_id (FK)
 - leave_type_id (FK)
 - start_date
 - end_date
 - reason
 - status (pending, approved, rejected)
 - approved_by (FK → employees.id)
 - created_at
-

6. Payroll Module

6.1 Salary Structure

salaries

- id (UUID, PK)
- employee_id (FK)
- base_salary
- bonus

- deductions
 - effective_from
-

6.2 Payroll Records

payroll

- id (UUID, PK)
 - employee_id (FK)
 - salary_id (FK)
 - month
 - year
 - gross_salary
 - net_salary
 - paid_at
-

7. Supporting Tables

7.1 Performance Reviews

performance_reviews

- id (UUID, PK)
 - employee_id (FK)
 - reviewer_id (FK → employees.id)
 - rating
 - comments
 - review_date
-

7.2 Documents

documents

- id (UUID, PK)
- employee_id (FK)
- file_url
- document_type (resume, contract, ID)
- uploaded_at

8. Relationships (Important)

- employees → departments (many-to-one)
- employees → positions (many-to-one)
- employees → employees (manager hierarchy)
- attendance → employees (many-to-one)
- leave_requests → employees (many-to-one)
- leave_requests → leave_types (many-to-one)
- payroll → employees (many-to-one)
- salaries → employees (one-to-many over time)